Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 2: Nutrition in Animals

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	2 — Nutrition in Animals
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic P4 traps: confusing the LOCATION of a step (gut cavity) with the step itself, mistaking emulsification for chemical digestion, treating absorption and assimilation as the same event, swapping rumen/abomasum and chyme/cud, and assuming the longest organ does the most of every job. Do not write on this paper — mark answers on your answer sheet.

---

1. A scientist writes the five steps of nutrition on cards and shuffles them. She gives one clue: 'Each step can only happen after the material it needs has been produced by the step before it.' Using only that rule, which TWO steps could a student be most certain about placing first and last?

(A) Digestion must be first and assimilation must be last
(B) Ingestion must be first and assimilation must be last
(C) Ingestion must be first and egestion must be last
(D) Absorption must be first and egestion must be last

2. A person's gut takes in food, breaks it down, and passes simple molecules into the blood, but a defect means the molecules circulate in the blood and are NEVER built into body tissue or burnt for energy. Over months the person eats normally yet wastes away. Which single step is failing, and what proves the three steps before it all worked?

(A) Absorption is failing; the wasting away proves digestion worked but nutrients never reached the blood
(B) Assimilation is failing; the molecules reaching the blood proves ingestion, digestion and absorption all worked

(C) Digestion is failing; the simple molecules in the blood prove ingestion worked but nothing was broken down

(D) Egestion is failing; the wasting away proves the body is keeping undigested waste inside

3. A student is told that faeces contain not only undigested fibre but also dead cells shed from the gut lining. She reasons: 'Since dead body cells are leaving, passing faeces must be excretion, not egestion.' Why is she wrong?

(A) Passing faeces is still egestion: the fibre and shed lining cells leave the food canal without ever having been taken into the body's living cells and used, whereas excretion removes waste the cells themselves made by their own chemistry, such as carbon dioxide

(B) She is right; because dead cells are present, the process must be excretion

(C) It is excretion, because the dead cells were once part of living body tissue

(D) It is neither egestion nor excretion, since a mix of food residue and cells fits no single term

4. A teacher says: 'Egestion and excretion both get rid of unwanted matter, yet only one of them ever involved the digestive enzymes.' Which is which, and why?

(A) Excretion involved the enzymes, since cell waste is made by enzymes; egestion never met any enzyme

(B) Egestion involved the digestive enzymes (they acted on the food whose residue is egested); excretion did not, as it removes waste made inside cells

(C) Both involved the enzymes, because all waste passes through the digested food

(D) Neither involved the enzymes, because enzymes are recycled and never reach waste

5. Following one mouthful of dal-rice: starch turns to sugar, sugar crosses into blood, blood carries sugar to a muscle cell, the cell burns it for energy. List, in order, the FOUR steps of nutrition these four events belong to.

(A) Digestion, absorption, assimilation, egestion

(B) Ingestion, digestion, absorption, assimilation

(C) Digestion, absorption, (transport is not a step), assimilation

(D) Digestion, assimilation, absorption, egestion

6. A student argues: 'If a food molecule is absorbed into my blood, then by definition my body has been nourished by it.' Choose the most precise correction.

(A) Correct — once a molecule is in the blood, assimilation has automatically happened

(B) Wrong — a molecule must be egested before the body counts as nourished

(C) Wrong — nourishment happens at digestion, before the molecule ever reaches the blood

(D) Not necessarily — absorption only puts the molecule into the blood; the body is nourished only after assimilation, when cells actually use it

7. An Amoeba, a snake, a butterfly and a housefly are filmed feeding. A judge must group them by the step of nutrition each clip shows. He notices all four clips, despite looking utterly different, capture the SAME step. Which step, and what is the common feature that overrides the differences?

(A) Digestion — every clip shows food being chemically broken down as it enters

(B) Absorption — every clip shows nutrients passing directly into body fluids at the point of feeding

(C) Egestion — every clip shows material being released from the organism

(D) Ingestion — every clip shows food being taken into the organism, regardless of the very different methods used

8. The housefly is special: it releases digestive juice onto solid food OUTSIDE its body, then sucks up the liquid. A student concludes 'the housefly therefore skips the digestion step entirely.' What is the accurate statement?

(A) The student is right; sucking liquid food means no digestion is ever needed in the housefly

(B) The housefly skips ingestion instead, because the food is already liquid when taken in

(C) The housefly carries out absorption before ingestion, since the juice absorbs nutrients outside the body

(D) The housefly does not skip digestion; it simply begins part of its digestion outside the body before completing ingestion

9. An Amoeba flows around a food particle and traps it. Years later the same Amoeba uses an identical kind of extension to creep toward a new particle. A biologist says this reveals an efficiency a human stomach does not have. What efficiency?

(A) The pseudopodium digests the food while also absorbing it, doing two chemical jobs at once

(B) One temporary structure (the pseudopodium) serves two jobs — both moving the cell and capturing food — instead of a fixed organ for each task

(C) The pseudopodium permanently stores the food so the Amoeba never needs to feed again

(D) The pseudopodium produces its own bile, letting the Amoeba digest fat without a liver

10. A mosquito pierces and sucks, a cow grinds with broad teeth, a python swallows prey whole, and an Amoeba engulfs with pseudopodia. A student claims these prove that 'animals differ in their step of digestion.' Identify the precise error.

(A) These differences are in INGESTION, not digestion; how food is taken in varies, but each is still the ingestion step

(B) The differences are in absorption, since each animal moves nutrients into blood differently

(C) The student is correct; the four animals do digest food in four different steps

(D) The differences are in assimilation, since each animal uses nutrients in its own way

11. A butterfly's long coiled tube and a snake's wide unhinging jaw are shaped utterly differently, yet a textbook places them in one column titled by a single nutrition step. The column's title is best completed as 'Tools for ___ , shaped by the kind of food eaten.'

(A) assimilation
(B) absorption
(C) egestion
(D) ingestion

12. An adult has 32 teeth: incisors and canines together number 12, and there are exactly three times as many grinding teeth as cutting-and-tearing teeth would suggest is impossible — so reconsider. Given 8 incisors, 4 canines, 8 premolars and 12 molars, what fraction of all the teeth can grind food?

(A) 12 out of 32, which is 3/8 of the teeth
(B) 20 out of 32, which is 5/8 of the teeth
(C) 8 out of 32, which is 1/4 of the teeth
(D) 32 out of 32, which is all of the teeth

13. A dentist's drill passes through the hard outer shell, then a softer middle layer, then strikes a region full of nerves and blood vessels, making the patient feel sharp pain. Name the three regions in the order the drill met them.

(A) Dentine, then enamel, then the pulp cavity

(B) Enamel, then dentine, then the pulp cavity

(C) Enamel, then pulp cavity, then dentine
(D) Pulp cavity, then dentine, then enamel

14. Two cavities are described. Cavity A has eaten through the enamel and into the dentine but the patient feels nothing. Cavity B is shallower but the patient feels sharp pain. The only way both descriptions can be true is if:

(A) Cavity A is painless because enamel contains the nerves, and Cavity B hurts because it scraped the enamel

(B) Cavity A, though deeper into dentine, has not yet reached the nerves in the pulp, while Cavity B has reached or nearly reached the pulp despite being narrower

(C) Cavity A is painless because dentine is the hardest layer, and Cavity B hurts because it is in the soft enamel

(D) Both descriptions cannot be true; a deeper cavity must always hurt more than a shallower one

15. Tooth decay follows this chain: leftover sugar -> mouth bacteria feed on it -> bacteria release acid -> acid dissolves enamel. A child who cannot brush after lunch wants the single most effective break in this chain. Which action breaks the chain EARLIEST?

(A) Chewing harder food afterwards so the enamel becomes tougher against the acid

(B) Eating a sweet dessert so the bacteria become full and stop making acid

(C) Swallowing an antacid tablet so the stomach acid does not rise to the teeth

(D) Rinsing the mouth with water to wash away the leftover sugar before bacteria can feed on it

16. A grass-eating animal and a flesh-eating animal are compared. The grass-eater has many broad, ridged back teeth and small front teeth; the flesh-eater has prominent pointed teeth beside its front teeth and fewer flat ones. From teeth alone, what can be most safely concluded?

(A) The flesh-eater is the larger animal, since pointed teeth only grow in big animals

(B) Both animals have identical sets of teeth; only their jaw muscles differ

(C) Tooth form reflects diet: broad ridged teeth suit grinding plant food, prominent pointed teeth suit tearing flesh

(D) Tooth shape is random and tells us nothing reliable about what an animal eats

17. If every one of a person's molars and premolars were lost but all incisors and canines remained, which everyday eating task would become MOST difficult, based purely on what each tooth type does?

(A) Biting off a piece of an apple, because the cutting teeth are gone

(B) Grinding tough or fibrous food into a fine pulp, because the grinding teeth are gone

(C) Tearing a piece of meat from the bone, because the tearing teeth are gone

(D) Swallowing liquids, because teeth are essential for swallowing

18. A student chews a starchy cracker for a long time and it turns sweet, then chews a piece of cheese (mostly fat and protein) for the same time and notices no sweetness. What does the contrast prove about saliva?

(A) Saliva's enzyme acts on starch (turning it to sugar) but not on the fat or protein in cheese

(B) Saliva digests all foods equally, so the cheese must secretly have turned sweet too

(C) Saliva acts only on protein, which is why neither food became sweet

(D) Saliva acts on fat, which is why the cheese should have tasted sweeter than the cracker

19. Saliva is said to have two distinct helps for a dry biscuit. A student lists only 'it starts digesting the biscuit.' Which second help is being overlooked, and why does swallowing depend on it?

(A) Saliva absorbs the biscuit's sugar into the blood right in the mouth, easing the load downstream

(B) Saliva makes hydrochloric acid in the mouth to soften the biscuit before swallowing

(C) Saliva moistens and softens the dry biscuit into a slippery lump, without which it could not slide down the food pipe easily

(D) Saliva cools the biscuit so it does not burn the food pipe on the way down

20. A laboratory keeps two warm tubes of starch: Tube 1 gets saliva, Tube 2 gets boiled-then-cooled saliva. After half an hour only Tube 1 shows sugar. Boiling is known to destroy enzymes. What is the strongest conclusion?

(A) Warmth alone makes starch into sugar, since both tubes were warm

(B) Saliva is acidic and the acid, not an enzyme, broke the starch in Tube 1

(C) The sugar in Tube 1 was produced by an enzyme in fresh saliva that boiling destroyed in Tube 2

(D) Boiled saliva works better, so Tube 2 should actually have shown more sugar

21. A person swallows while hanging upside-down and the water still reaches the stomach. A friend insists 'this only works because some water leaks down by gravity.' Refute this with the strongest single point.

(A) Gravity is pulling water AWAY from the stomach here, so the water reaching the stomach must be pushed by peristaltic muscle waves, not pulled by gravity

(B) Gravity still helps, because water is heavy and always finds the stomach

(C) The food pipe is a straight vertical tube, so position does not matter

(D) Water is absorbed by the food pipe directly, so it never needs to travel to the stomach

22. A medicine freezes the muscles of the food pipe so they can no longer contract in waves, but leaves the stomach, intestines and saliva working. A person tries to eat a sandwich. What is the most direct problem?

(A) The stomach will fail to churn, so the sandwich rots in the food pipe

(B) The chewed sandwich, though swallowed, cannot be pushed down the food pipe to the stomach, so it gets stuck

(C) Salivary amylase will stop working, so the sandwich tastes bitter

(D) Water will be over-absorbed in the food pipe, drying the sandwich into a hard lump

23. Order three locations by HOW LONG food typically stays there: a wave-pushing tube it passes through in seconds, a churning bag it stays in for a few hours, and a long coiled tube it stays in for many hours. Name them shortest-stay to longest-stay.

(A) Oesophagus (seconds), stomach (a few hours), small intestine (many hours)

(B) Stomach (seconds), oesophagus (a few hours), small intestine (many hours)

(C) Oesophagus (seconds), small intestine (a few hours), stomach (many hours)

(D) Small intestine (seconds), stomach (a few hours), oesophagus (many hours)

24. Gastric juice has three components: hydrochloric acid, mucus and pepsin. A student assigns each a job and gets ONE wrong. Which assignment is the error?

(A) 'Hydrochloric acid kills germs in the food' is the error

(B) 'Pepsin begins digesting proteins' is the error

(C) 'Mucus digests proteins' is the error — mucus protects the wall, while pepsin digests proteins

(D) 'Hydrochloric acid makes the surroundings acidic for pepsin' is the error

25. A patient produces almost no hydrochloric acid in the stomach. Two harmful effects are expected. Which pair correctly names them, tracing each from what the acid normally does?

(A) Pepsin works poorly so stomach protein digestion is impaired, and swallowed germs are less effectively killed

(B) Starch digestion fails everywhere, and bile can no longer reach the small intestine

(C) Fat is no longer emulsified, and water absorption in the large intestine stops

(D) Saliva stops being made, and the food pipe loses its peristalsis

26. Surgeons note that if the protective lining of the stomach is stripped away in just one spot, an ulcer (a self-digested sore) forms there, even though the rest of the wall — equally made of protein — stays intact. What does the ulcer appearing only at the stripped spot prove about how the stomach normally avoids digesting itself?

(A) It proves the stomach wall contains no protein except at the stripped spot

(B) It proves bile from the liver normally coats and protects the whole wall

(C) It proves the acid and pepsin are dangerous only where the wall is thickest

(D) It proves the wall IS digestible protein, and the normal protection is the mucus layer coating the lining — remove that layer and the exposed protein is attacked

27. In a patient, the fluid that normally neutralises stomach acid fails to reach the first part of the small intestine, so the chyme there stays as strongly acidic as it was in the stomach. Pancreatic enzymes are present in normal amounts. Predict the effect on digestion in that region, and why.

(A) Digestion speeds up there, because more acid always means faster enzyme action

(B) Only protein digestion improves there, because pepsin keeps working in the acid

(C) Nothing changes, because intestinal enzymes work equally well at any acidity

(D) Digestion there is impaired, because the pancreatic enzymes need non-acidic (neutralised) surroundings and cannot work well while the chyme stays strongly acidic

28. A quiz gives two clues: Material 1 is made inside an animal with a single-chambered stomach and is never chewed again; Material 2 is brought back up to the mouth and chewed a second time. A student must name each. Which naming is correct?

(A) Material 1 is cud and Material 2 is chyme

(B) Both are chyme, since both are partly digested food

(C) Both are cud, since 'cud' covers any partly digested mass

(D) Material 1 is chyme (the single-stomach paste, never re-chewed) and Material 2 is cud (regurgitated and chewed again)

29. Three organs are described only by what they do to food: Organ X churns and mixes food with acid, Organ Y completes digestion and absorbs most nutrients, Organ Z recovers water from the leftover. Place them in the order food actually meets them.

(A) Y (small intestine) first, then X (stomach), then Z (large intestine)

(B) X (stomach) first, then Y (small intestine), then Z (large intestine)

(C) X (stomach) first, then Z (large intestine), then Y (small intestine)

(D) Z (large intestine) first, then Y (small intestine), then X (stomach)

30. After a heavy meal the stomach churns vigorously but absorbs almost no nutrients into the blood, while the small intestine absorbs a great deal. A student is puzzled because 'the stomach holds food longer at one time.' What resolves the puzzle?

(A) The stomach actually absorbs more; the student measured it wrongly

(B) The small intestine churns harder than the stomach, which is why it absorbs more

(C) The stomach's role is to churn and start digestion, while absorption of nutrients is mainly the small intestine's job, thanks to its villi

(D) The stomach absorbs nothing because it has no blood supply at all

31. An engineer is told that if the small intestine's millions of finger-like projections were flattened smooth, the wall would keep only a tiny fraction of its present absorbing surface. Using only this fact, what is the single best explanation for why the projections matter to nutrition?

(A) The projections make food travel faster, leaving less waste behind

(B) The projections pack a very large absorbing surface into a limited length of gut, so far more digested nutrients can be taken up

(C) The projections make the intestine more acidic, dissolving nutrients faster

(D) The projections manufacture extra bile to finish digesting fats

32. A child with flattened intestinal villi is undernourished despite eating well. After a year on a special diet the villi regrow and become finger-like again, and the child's weight returns to normal even though the amount eaten does not change. What does the recovery demonstrate?

(A) Eating more food, not the villi, controls nourishment

(B) The stomach, not the villi, absorbs the nutrients once it churns harder

(C) Nourishment depended on the villi providing absorbing surface: lose them and absorption falls, regrow them and absorption — and weight — recover

(D) Bile production rose with the new diet and absorbed the nutrients directly

33. After a protein meal, an amino acid crosses the wall of a villus into the blood; weeks later that same amino acid is found built into a brand-new muscle fibre the child has grown. Name the nutrition step at the villus and the nutrition step in the growing muscle.

(A) Digestion at the villus, then absorption in the muscle

(B) Absorption at the villus, then assimilation in the muscle

(C) Assimilation at the villus, then absorption in the muscle

(D) Absorption at the villus, then egestion in the muscle

34. A student claims 'the villi must be where digestion is completed, because that is where the simple molecules finally appear.' Correct the reasoning precisely.

(A) The student is right; villi are the main digesting organs of the body

(B) The villi both digest and store the molecules, which is why they appear there

(C) The molecules appear at the villi because the villi churn the food like a small stomach

(D) The simple molecules appear at the villi only because they are absorbed there; the digestion that made them happened in the gut cavity before reaching the villi

35. Comparing two events in the small intestine: a fat globule being broken into tiny droplets, and an amino acid crossing into the blood. A student says 'both are digestion.' Identify the correct labels.

(A) Both are absorption, since both move material across the intestinal wall

(B) Both are digestion, since both involve breaking molecules apart

(C) Breaking the fat into droplets is part of digestion (preparing fat for enzymes), but the amino acid crossing into blood is absorption, not digestion

(D) Breaking the fat into droplets is absorption, and the amino acid crossing is digestion

36. The large intestine is much shorter than the small intestine yet a student insists 'longer organs always do the more important nutrient work, so the large intestine must absorb most nutrients.' Why is this doubly wrong?

(A) Length does not decide importance; the small intestine absorbs most nutrients, and the large intestine's job is recovering water, not absorbing nutrients

(B) It is right; the large intestine, being last, absorbs all the nutrients the small intestine missed

(C) It is wrong only about length; the large intestine does absorb most nutrients despite being short

(D) It is wrong only about importance; the large intestine still absorbs most nutrients

37. After an infection rushes food through the gut so fast that the large intestine has almost no time to take water back from the residue, the person passes very loose, watery stools. Using this, predict what passing hard, dry stools (constipation) would instead indicate about water in the large intestine.

(A) Constipation would mean the large intestine took back too little water, the same as the watery case

(B) Constipation would mean the large intestine stopped handling water altogether

(C) Constipation would mean stomach acid was being vomited up

(D) Constipation would mean the large intestine took back too MUCH water from the residue, the opposite of the watery-stool case

38. A doctor follows a swallowed metal coin on X-rays. It travels through the stomach and intestines unchanged and finally leaves with the faeces. Naming the five steps, which steps did the coin take part in, and which could it never take part in?

(A) It was digested and absorbed, then egested as nutrients

(B) It was assimilated into the child's bones along with the meal's calcium

(C) It was excreted as a chemical waste made inside the body's cells

(D) It took part only in ingestion (it was swallowed) and egestion (it left with the residue); being unbreakable, it could never be digested, absorbed or assimilated

39. In a tube, a large blob of oil sits in water; adding a drop of bile breaks the blob into a cloud of tiny droplets, yet a chemical test shows the oil's molecules are still whole, not cut into smaller ones. A student says this proves bile 'digests' the fat. Correct the statement and name what bile actually did and where it does so in the body.

(A) The student is right; breaking the blob apart is chemical digestion of the fat

(B) Bile chemically cut the oil into amino acids, which the test missed

(C) Bile did NOT digest the fat (the molecules are unchanged); it emulsified it — physically breaking large fat globules into tiny droplets — which in the body happens in the small intestine and lets fat-digesting enzymes work faster

(D) Bile turned the oil into starch, which the test cannot detect

40. A person whose bile-storing sac was removed is advised to avoid large fatty meals, but is told starchy and protein meals need no special caution. From this advice alone, what can be concluded about the fluid that sac stored and the food it chiefly helps with?

(A) The sac stored pepsin, the stomach's fat-digesting enzyme

(B) The sac stored bile, and bile chiefly helps with fats — which is why only large fatty meals need caution

(C) The sac stored amylase, so starchy meals should have needed the caution instead

(D) The sac made stomach acid, so all three food types should need equal caution

41. A bile-duct blockage stops bile from reaching the small intestine, while saliva, gastric juice and pancreatic enzymes all still arrive. Of starch, protein and fat, which is MOST hampered, and what subtle effect remains on the others?

(A) Fat is most hampered because, without bile, it stays in large globules that enzymes struggle to act on; starch and protein digestion continue almost normally

(B) Starch is most hampered, because bile is the main starch enzyme; fat is unaffected

(C) Protein is most hampered, because bile activates the protein enzymes; fat is unaffected

(D) All three are hampered equally, because bile is required to digest every food type

42. A table claims, for one secretion, both the food it acts on and the nature of its action. Which single row is correct in BOTH columns?

(A) Gastric pepsin — acts on starch — by chemical breakdown

(B) Saliva — acts on fat — by emulsification

(C) Bile — acts on protein — by chemical breakdown

(D) Bile — acts on fat — by emulsification (a physical change, with no enzyme)

43. Three students each predict the result of blocking ONLY the pancreatic juice while saliva and gastric juice still flow. Pancreatic juice carries enzymes for carbohydrates, proteins and fats, all acting in the small intestine. Which prediction is correct?

(A) Only mouth starch digestion fails, since the pancreas makes salivary amylase

(B) Only stomach protein digestion fails, since the pancreas makes pepsin

(C) Only large-intestine water absorption fails, since the pancreas absorbs water

(D) Carbohydrate, protein and fat digestion in the small intestine are all impaired together, since the pancreas supplies the small intestine's enzymes for all three

44. Why must the strongly acidic chyme leaving the stomach be neutralised before the pancreatic enzymes act on it in the small intestine?

(A) Because the pancreatic enzymes work in non-acidic conditions, so the acid from the stomach must first be neutralised for them to function

(B) Because the pancreatic enzymes need even stronger acid than pepsin to work

(C) Because neutralising the acid is what actually digests the carbohydrates and fats

(D) Because the small intestine cannot hold any liquid that is not perfectly dry

45. A scientist jokes that a human could 'live like a cow on grass' only if given two things the cow's body provides for itself. Grass is mostly cellulose, which no mammal's own enzymes break down. What are the two things, and why is each needed?

(A) A bigger stomach and more teeth, to hold and grind more grass

(B) More saliva and more stomach acid, to dissolve the cellulose chemically

(C) Cellulose-digesting microbes (as live in a cow's rumen) and the cud-re-chewing that grinds grass finer for those microbes, because the human's own enzymes cannot break cellulose

(D) A longer food pipe and a gall bladder, to push and emulsify the grass

46. In a ruminant, Chamber 1 is the large fermentation vat where microbes break down cellulose; Chamber 2 works like a human stomach, using acid and enzymes. A student must name them, avoiding the usual mix-up. Which naming is right?

(A) Chamber 1 is the abomasum; Chamber 2 is the rumen

(B) Chamber 1 is the rumen; Chamber 2 is the abomasum

(C) Chamber 1 is the rumen; Chamber 2 is the reticulum

(D) Chamber 1 is the omasum; Chamber 2 is the rumen

47. A cow eats grass quickly in the morning, then rests in the afternoon chewing material brought back up from its stomach. A student says 'the cow is eating fresh grass again.' What is actually happening, and what is this re-chewed material called?

(A) The cow is eating fresh grass a second time, which is why it chews twice a day

(B) The cow is chewing chyme, the same paste a human stomach makes, brought up to the mouth

(C) The cow is chewing grass that grew inside its stomach during the morning

(D) The cow is re-chewing the cud — partly digested grass brought back up from the stomach — not eating new grass from outside

48. A cow with an injury can no longer bring food back up to re-chew, though its rumen microbes stay healthy. Fed the same grass, it now extracts noticeably LESS energy from it than before. What chain of cause and effect explains the drop?

(A) Without re-chewing, the grass is heated less, and heat is what releases the energy

(B) Without re-chewing, the grass stays in coarser pieces with less exposed surface, so the rumen microbes break down less of its cellulose and free fewer usable nutrients

(C) Without re-chewing, the grass turns to protein in the mouth instead of to energy

(D) Without re-chewing, the cellulose is absorbed whole in the mouth and wasted

49. A goat chews cud for long minutes, yet measurements show the breakdown of cellulose into usable substances keeps rising for hours AFTER the chewing stops, while the cud sits in the largest stomach chamber. What does this timing reveal about where cellulose is really broken down?

(A) It is broken down in the mouth; the long chewing does all the work

(B) The mouth only grinds the cud finer; the cellulose itself is broken down later by microbes in the rumen, which is why breakdown continues for hours after chewing stops

(C) It is broken down in the food pipe as the cud is re-swallowed

(D) It is never broken down; the goat merely stores the cud unchanged forever

50. A cow is grouped with deer and buffalo as a ruminant, while a human is not. A student lists four reasons; three are false. Which is the ONE true defining reason?

(A) Ruminants swallow plant food quickly and later return it to the mouth to chew again as cud

(B) Ruminants eat meat, which is why they need extra stomach chambers

(C) Ruminants digest all their food inside a single cell, unlike humans

(D) Ruminants have only one stomach chamber, the same as a human

51. An Amoeba traps food in a bubble-like pocket, digests and absorbs inside it, then moves the pocket to the cell edge and dumps the leftover outside. A student maps these onto human terms and must pick the right pair for 'the pocket' and 'the dumping.'

(A) The pocket is a gall bladder; the dumping is excretion of cell-made waste

(B) The pocket is a villus; the dumping is absorption of the leftover

(C) The pocket is a rumen; the dumping is assimilation of the leftover

(D) The pocket is a food vacuole; the dumping of leftover is egestion, as the vacuole fuses with the cell surface

52. A biologist stresses that in BOTH an Amoeba and a human, digestive enzymes never act inside the body's own working cells nor in the blood, but in an enclosed pocket or tube. For the human, the place that matches the Amoeba's food vacuole is:

(A) The hollow cavity of the stomach and intestine — enclosed spaces, but still outside the body's own cells and outside the blood

(B) Inside the blood, where enzymes act on nutrients once they are absorbed

(C) Inside the muscle cells, where the nutrients are finally burnt for energy

(D) On the outer skin surface, before the food is even swallowed

53. An Amoeba forms a pseudopodium only when it needs to move or feed, then withdraws it; a human keeps a permanent stomach, intestine and so on at all times. Setting aside which is 'better,' what core body-plan fact does this contrast show?

(A) The Amoeba keeps a separate permanent organ for each step, only smaller than a human's

(B) The Amoeba digests entirely outside its body, so it needs no internal space at all

(C) Being a single cell, the Amoeba makes temporary structures on demand for different jobs, instead of maintaining the fixed, specialised organs a multi-organ human relies on

(D) The Amoeba cannot move, so the pseudopodium is only for sensing its surroundings

54. A teacher claims: 'An Amoeba and a human both complete all five steps of nutrition; the difference is only in the body plan that carries them out.' Choose the statement that both supports the claim AND correctly states the difference.

(A) Both ingest, digest, absorb, assimilate and egest; but the Amoeba does all five within one cell while the human spreads them across many specialised organs

(B) Only the human truly digests food; the Amoeba merely engulfs and stores it without digestion

(C) The Amoeba absorbs nutrients but never egests, so it completes only four of the five steps

(D) The human ingests and digests but never assimilates, so neither truly completes all five

55. A poison stops the tiny cell structures where glucose is 'burnt' to release energy from working, even though glucose still reaches the cells normally. Name those structures, and name the nutrition step that this energy-releasing USE of glucose belongs to.

(A) The mitochondria; and the use of absorbed glucose this way is part of assimilation

(B) The villi; and this energy use is part of absorption

(C) The food vacuoles; and this energy use is part of digestion

(D) The salivary glands; and this energy use is part of ingestion

56. Put the whole pathway together for a meal of paneer (protein-rich): name where its protein digestion BEGINS, where that digestion is COMPLETED, and where its building blocks are ABSORBED.

(A) Begins in the mouth, completed in the stomach, absorbed in the large intestine

(B) Begins in the small intestine, completed in the stomach, absorbed in the mouth

(C) Begins in the stomach, completed in the small intestine, and the building blocks are absorbed in the small intestine

(D) Begins in the large intestine, completed in the mouth, absorbed in the stomach

57. A single meal of buttered toast contains starch (toast) and fat (butter). Name the organ where the starch's digestion FIRST begins and the organ where the butter's fat is MAINLY emulsified and digested.

(A) Starch digestion first begins in the mouth; the fat is mainly emulsified and digested in the small intestine

(B) Both the starch and the fat first begin digestion in the stomach

(C) Starch digestion first begins in the small intestine; the fat is digested in the mouth

(D) Both the starch and the fat are digested only in the large intestine

58. An athlete gulps a sugary sports drink (dissolved sugar in water). Track where the dissolved SUGAR is mainly taken into the blood versus where most of the remaining WATER is finally reabsorbed. The correct pairing is:

(A) Both the sugar and the water are mainly taken up in the stomach

(B) Sugar is mainly taken up in the large intestine; water in the small intestine

(C) Sugar is mainly taken up in the small intestine (across the villi); most remaining water is reabsorbed in the large intestine

(D) Both the sugar and the water are mainly taken up in the mouth

59. An Amoeba wraps a particle in pseudopodia, a butterfly draws nectar through a coiled tube, and a python swallows prey whole. A student insists these three differ in their METHOD of digestion. Pinpoint the conceptual mistake.

(A) They differ in absorption, since each passes nutrients into its body fluids by a different route

(B) The student is correct; engulfing, sipping and swallowing are three kinds of digestion

(C) The three differ in their method of ingestion (how food is taken in), not digestion; digestion is the later chemical breakdown, which the clips do not show differing

(D) They differ in assimilation, since each builds body tissue from food in its own way

60. Trace a single sip of milk (containing fat, protein and milk-sugar). At the moment it first enters the mouth, exactly how many of the five steps of nutrition have been completed for it, and which one is occurring?

(A) Zero steps are completed; the step occurring at that moment is ingestion (taking the milk in)

(B) One step is completed (digestion) and absorption is now occurring

(C) Two steps are completed (ingestion and digestion) and assimilation is now occurring

(D) Four steps are completed and only egestion remains

Solutions — Science (Class 7), Chapter 2: Nutrition in Animals — Paper 4

Paper 4 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	D	21	A	31	B	41	A	51	D
2	B	12	B	22	B	32	C	42	D	52	A
3	A	13	B	23	A	33	B	43	D	53	C
4	B	14	B	24	C	34	D	44	A	54	A
5	C	15	D	25	A	35	C	45	C	55	A
6	D	16	C	26	D	36	A	46	B	56	C
7	D	17	B	27	D	37	D	47	D	57	A
8	D	18	A	28	D	38	D	48	B	58	C
9	B	19	C	29	B	39	C	49	B	59	C
10	A	20	C	30	C	40	B	50	A	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Nothing can happen until food is taken in, so ingestion is necessarily first. Egestion removes the residue left after digestion and absorption have taken what is useful, so it can only be last. Digestion cannot be first because it needs food already ingested. Assimilation is not last — egestion of undigested residue follows it, since the leftover never entered cells. Absorption cannot be first because there is nothing simple to absorb until digestion has occurred.

2. (B) Simple molecules circulating in the blood prove the food was taken in (ingestion), broken down (digestion) and passed into the blood (absorption). The failure is that cells cannot USE them for building or energy — that is assimilation. Absorption cannot be failing because the molecules clearly reached the blood. Digestion cannot be failing because simple molecules exist. Egestion concerns undigested waste leaving the body, not the use of absorbed nutrients.

3. (A) Egestion is defined by the SOURCE of the material — undigested matter that passed along the food canal without ever entering the body's living cells — not by whether a few shed lining cells are mixed in. Those dead cells are simply carried out with the residue, so passing faeces remains egestion. Excretion, by contrast, removes waste the cells themselves made by their own chemistry, such as carbon dioxide. So (B) and (C) wrongly let stray dead cells redefine the step; (D) invents a non-category — the process clearly fits egestion.

4. (B) The matter that is egested is the residue of food that the digestive enzymes worked on but could not break down (e.g. fibre). So the egested material was exposed to digestive enzymes in the canal. Excreted waste — like carbon dioxide from energy release — is produced inside cells far from the digestive enzymes, so it never met them. Saying excretion involved digestive enzymes confuses metabolic chemistry inside cells with digestion. Both did not meet enzymes, because cell waste forms separately.

5. (C) Starch-to-sugar is digestion. Sugar crossing into blood is absorption. Blood merely carrying sugar is transport, which is NOT one of the five named steps. The cell burning sugar for energy is assimilation (using absorbed nutrients). So the steps named, in order, are digestion, absorption, then assimilation, with transport sitting between absorption and assimilation but unnamed. Adding egestion is wrong — nothing is being thrown out here. Adding ingestion is wrong — the food was already in the mouth before this chain begins. Putting assimilation before absorption reverses cause and effect.

6. (D) Absorption merely transfers nutrients from the gut into the blood. The body is truly nourished only when cells take up and USE those nutrients — that is assimilation, a later, separate step. So absorption alone does not guarantee nourishment (a cell defect could leave the nutrient unused, as in question 2). Egestion is the removal of undigested residue and has nothing to do with nourishment. Digestion only breaks food into absorbable units; it is not yet nourishment either.

7. (D) Engulfing by pseudopodia (Amoeba), swallowing prey whole (snake), sucking nectar (butterfly) and lapping liquefied food (housefly) are all just methods of TAKING FOOD IN — that is ingestion. The differing tools and forms of food are the variations; the common act of taking food into the body is the shared step. It is not digestion — chemical breakdown follows ingestion and is not what these clips primarily show. It is not absorption — none of these acts moves nutrients into body fluids at the feeding point. It is not egestion — food is going in, not out.

8. (D) Pouring digestive juice onto food and dissolving it IS digestion — it has merely been moved to before/around ingestion rather than skipped. So the student is wrong that digestion is skipped. Ingestion still occurs: the housefly sucks the liquid in, which is taking food into the body. Absorption (nutrients into body fluids) cannot occur outside the body before the food has even been taken in; the external juice only digests, it does not absorb.

9. (B) A pseudopodium is a temporary cytoplasmic extension used both for locomotion and for engulfing food, so a single improvised structure does two jobs — an economy a fixed organ such as the human stomach lacks. It does not both digest and absorb at once; digestion happens later inside the food vacuole. It does not store food permanently — leftovers are egested. Amoeba has no bile or liver, so it does not make bile.

10. (A) Piercing-and-sucking, grinding, whole-swallowing and engulfing are all different METHODS OF TAKING FOOD IN — they differ in ingestion, not digestion. The student mislabels the step. Absorption (nutrients into blood) is an internal step and is not what distinguishes these feeding methods. Assimilation is the cellular use of nutrients, far downstream of how food enters, so it is not what differs here either.

11. (D) Both the butterfly's proboscis and the snake's expandable jaw are devices for TAKING FOOD IN, so the shared step is ingestion, and the differing shapes reflect different foods (liquid nectar versus whole prey). Assimilation happens inside cells and uses no feeding tool. Absorption occurs at the gut wall, not at these mouthparts. Egestion is waste removal, unrelated to mouthpart shape.

12. (B) Grinding teeth are the premolars (8) and molars (12), totalling 20. As a fraction of 32 teeth that is 20/32 = 5/8. Choosing 12 counts only the molars and ignores the 8 premolars that also grind. Choosing 8 counts only the premolars and ignores the molars. Choosing 32 wrongly treats every tooth as a grinder, ignoring the 12 incisors and canines which cut and tear.

13. (B) From outside inward a tooth crown is enamel (hardest outer shell), then dentine (softer middle layer), then the pulp cavity (centre, with nerves and blood vessels — hence the pain when reached). The first option correctly orders them. Putting dentine before enamel reverses the outermost two. Putting the pulp before the dentine skips the middle layer. Starting at the pulp reverses the whole sequence — the drill enters from the outside, not the centre.

14. (B) Pain comes from reaching the nerves in the pulp cavity. Depth measured as 'how far through enamel and dentine' is not the same as 'how close to the pulp', because cavities can be wide-but-shallow or narrow-but-deep. Cavity A reached dentine but not the pulp, so no pain; Cavity B is narrower yet has reached or nearly reached the pulp, so it hurts. Enamel does not contain nerves, so a cavity in enamel alone would not hurt. Dentine is not the hardest layer (enamel is), and enamel is not soft. The 'cannot be true' option ignores that pain depends on proximity to the pulp, not total tissue removed.

15. (D) The chain begins with leftover sugar feeding bacteria. Removing that sugar (e.g. by rinsing) cuts the chain at its very first link, before bacteria even feed. Chewing harder food does not toughen enamel and acts only at the last link. Eating more sweets adds MORE sugar, feeding the bacteria further — the opposite of breaking the chain. Stomach acid is not the cause of dental decay here; the acid in this chain is made by mouth bacteria, so an antacid does nothing about it.

16. (C) Broad ridged molars grind tough plant matter, while long pointed canines tear flesh — so teeth track diet, which is the safe conclusion. Pointed teeth are about tearing, not body size, so nothing about size can be concluded. The two sets are clearly not identical, since one has prominent pointed teeth and the other broad grinders. Tooth shape is far from random — it closely matches feeding habit, which is the whole point of the comparison.

17. (B) Premolars and molars do the grinding, so losing them makes grinding fibrous food into a pulp hardest. Biting off a piece of apple uses incisors (cutting), which remain. Tearing meat uses canines (tearing), which remain. Swallowing liquids needs no teeth at all, so it is unaffected. Only grinding depends on the lost teeth.

18. (A) The cracker's starch is broken by salivary amylase into sugar, tasting sweet; the cheese's fat and protein are not touched by salivary amylase, so no sweetness appears. This pinpoints that saliva's enzyme is specific to starch. Saliva does not digest all foods equally — the cheese result disproves that. It does not act mainly on protein (protein digestion starts in the stomach). It does not act on fat, so the cheese would not become sweet.

19. (C) Besides starting starch digestion, saliva wets and softens dry food into a smooth, slippery bolus that can be swallowed and pushed along the food pipe. Saliva does not absorb sugar into the blood — absorption happens in the small intestine, not the mouth. Hydrochloric acid is made in the stomach, not by salivary glands. Cooling food is not a recognised role of saliva and is not what swallowing depends on.

20. (C) Both tubes were warm and both had saliva, but only the tube with UNBOILED saliva made sugar; since boiling destroys enzymes, the active agent must be a heat-sensitive enzyme (salivary amylase). Warmth alone cannot be the cause, because both tubes were warm yet only one made sugar. Acid is not the agent — saliva's action on starch is enzymatic, and the only difference between tubes was boiling, which targets enzymes. Boiled saliva did not work better; it showed no sugar, the opposite of that claim.

21. (A) Upside-down, gravity pulls the swallowed water toward the head, AWAY from the stomach, yet it still arrives — so something must actively push it against gravity, namely peristalsis (rhythmic muscle contractions of the food-pipe wall). Saying gravity helps is contradicted by the upside-down geometry. The food pipe is not a straight vertical drop; even if it were, gravity would oppose the motion here. Water is not absorbed by the food pipe; it must travel to the stomach, so peristalsis is essential.

22. (B) Peristalsis is what moves swallowed food down the food pipe; freezing those muscle waves means the food cannot travel to the stomach and lodges in the pipe. The stomach is said to be working, so churning is unaffected. Salivary amylase is in the mouth, also unaffected by a food-pipe problem. The food pipe does not absorb water, so over-absorption there is not a recognised effect.

23. (A) Food merely passes through the oesophagus in seconds (it is a transport tube moved by peristalsis), stays a few hours in the churning stomach, and lingers many hours in the long coiled small intestine where digestion is completed and most nutrients are absorbed. The first option matches. The others mislabel the seconds-long tube as the stomach or the small intestine, or claim food rushes through the small intestine in seconds, which contradicts its role as the slow site of most absorption.

24. (C) Pepsin is the protein-digesting enzyme; mucus's job is to coat and protect the stomach lining from the acid and pepsin, not to digest protein. So 'mucus digests proteins' is the wrong assignment. Hydrochloric acid does kill germs and does create the acidic conditions pepsin needs — both are correct. Pepsin does begin protein digestion — also correct. Only the mucus claim is mistaken.

25. (A) Acid does two jobs: it creates the acidic conditions pepsin needs (so too little acid impairs stomach protein digestion) and it kills many swallowed germs (so too little acid lets more germs survive). Starch digestion uses amylase, not stomach acid, and bile flow is unrelated to acid. Fat emulsification is bile's job and water absorption is the large intestine's, neither depending on stomach acid. Saliva and peristalsis are upstream and unaffected by stomach acid levels.

26. (D) The ulcer forming exactly where the lining was stripped — while the equally protein-rich rest of the wall stays intact — proves the wall IS digestible protein and that the lining is what shields it. That shield is the mucus layer; remove it and the exposed protein is attacked by acid and pepsin, giving the sore. So (A) is wrong: the wall does contain protein everywhere, not only at the spot. (B) is wrong: bile acts in the small intestine, not on the stomach wall. (C) is wrong: the ulcer depends on loss of mucus, not on wall thickness.

27. (D) The pancreatic (intestinal) enzymes work in non-acidic surroundings; if the neutralising fluid never arrives, the chyme stays strongly acidic and those enzymes cannot work well, so digestion in that region is impaired — answer (D). (A) is wrong: more acid does not speed these enzymes; it disables them. (B) only helps pepsin, which is finishing anyway, and does not rescue the intestinal enzymes. (C) is false: the intestinal enzymes are sensitive to acidity, not indifferent to it.

28. (D) Chyme is the semi-liquid paste of a single-chambered (human-type) stomach and is never re-chewed; cud is the partly digested food a ruminant brings back up to chew a second time. Matching each clue to its defining property gives Material 1 = chyme, Material 2 = cud — answer (D). (A) reverses them. (B) is wrong: cud is specific to ruminant regurgitation, not just any partly digested food. (C) is wrong: the single-stomach paste is chyme, a distinct term.

29. (B) Food is churned with acid in the stomach (X), then most digestion and absorption happen in the small intestine (Y), and finally water is recovered from the residue in the large intestine (Z). So the order is X, Y, Z. Putting the small intestine before the stomach reverses the path. Putting the large intestine before the small intestine would recover water before nutrients are even absorbed. Running the whole sequence backwards contradicts the real journey of food.

30. (C) Holding food longer does not make an organ absorb more; what matters is its function and structure. The stomach is built to churn and begin digestion, whereas the small intestine, lined with absorptive villi, is the main site of nutrient absorption. The stomach does not absorb more than the small intestine. The small intestine does not out-churn the stomach; churning is the stomach's specialty. The stomach does have a blood supply; its low absorption is about its role, not a lack of blood.

31. (B) The fact that flattening the projections would leave only a tiny fraction of the surface tells us their whole value is the absorbing surface they add: they pack a very large absorbing area into a limited length, so far more digested nutrient can be taken up — answer (B). (A) is wrong: villi are about surface, not speed, and faster transit would mean less absorption. (C) is wrong: absorption is not driven by acidity. (D) is wrong: bile comes from the liver, not from the projections.

32. (C) The weight falls when the villi are flat and returns when they regrow — with the amount eaten held constant — so the controlling factor is the absorbing surface the villi provide: lose it and absorption (and weight) drop; regrow it and both recover. That is answer (C). (A) is wrong: eating did not change, yet weight did. (B) is wrong: the stomach's churning does not absorb nutrients. (D) is wrong: bile only emulsifies fat; it cannot absorb nutrients in place of villi.

33. (B) The amino acid crossing the villus wall into the blood is absorption; the growing muscle then BUILDING that amino acid into new tissue is assimilation (cells using absorbed nutrients to build and repair). So the order is absorption then assimilation — answer (B). Digestion is not happening at the villus; the amino acid is already a simple molecule. Assimilation cannot precede absorption, so (C) reverses the chain. The muscle is using the amino acid to grow, not throwing it out, so egestion (D) is wrong.

34. (D) By the time molecules reach the villi they are ALREADY simple — digestion was completed earlier in the gut cavity by enzymes. The villi merely absorb these ready-made simple molecules into the blood; they do not digest. So the student confuses the site of absorption with the site of digestion. Villi do not digest or store food, nor do they churn like a stomach — absorption is their one job.

35. (C) Emulsification (breaking fat into droplets) prepares fat for chemical breakdown, so it belongs to digestion. An amino acid crossing the gut wall into the blood is absorption, a different step. So they are not both digestion or both absorption. The fourth option swaps the two labels — emulsification is not absorption (nothing crosses into blood there) and an amino acid entering blood is not digestion.

36. (A) Two errors: first, organ length does not determine which does the key nutrient work; second, nutrient absorption is the small intestine's job (it is the longer one and is lined with villi), while the large intestine mainly recovers water. The other options each keep the false claim that the large intestine absorbs most nutrients, which contradicts its actual water-recovery role.

37. (D) The watery-stool case shows that too LITTLE water taken back leaves loose, watery residue. Constipation (hard, dry stools) is the opposite outcome, so it must come from taking back too MUCH water — answer (D). (A) is wrong: it would make both cases identical. (B) is wrong: the organ still handles water, just to a different degree. (C) is wrong: vomiting acid has nothing to do with large-intestine water recovery.

38. (D) The coin travels through unchanged and leaves with the faeces, so it took part only in ingestion (being swallowed) and egestion (leaving with the residue); being unbreakable, it was never digested, absorbed or assimilated — answer (D). (A) is wrong: nothing digested or absorbed the coin. (B) is wrong: assimilation needs absorbed nutrients, which the coin never becomes. (C) is wrong: excretion removes cell-made waste, not an object that merely passed through the gut.

39. (C) The chemical test shows the oil molecules are still whole, so bile did NOT digest the fat; it emulsified it — a physical breaking of large globules into tiny droplets, which in the body happens in the small intestine and gives fat-digesting enzymes far more surface to act on. That is answer (C). (A) is wrong: breaking a blob into droplets is physical, not chemical digestion. (B) is wrong: amino acids come from protein, and the molecules were unchanged anyway. (D) is wrong: fat cannot turn into starch.

40. (B) Only large fatty meals need caution, so the lost sac's stored fluid must mainly help fats — that fluid is bile, stored in the gall bladder. Answer (B). (A) is wrong: pepsin is a stomach protein enzyme, not stored in the sac. (C) is wrong: had it stored amylase, starchy meals would suffer, but they are fine. (D) is wrong: had it made stomach acid, all three food types would need caution, not just fats.

41. (A) Bile emulsifies fat into droplets; without it, fat remains in large globules and its enzymes work poorly, so fat digestion is most hampered. Starch (amylase) and protein (pepsin and pancreatic enzymes) do not need bile, so they proceed almost normally. Bile is not a starch enzyme, and amylase still arrives. Bile does not activate protein enzymes. Bile is not required for every food type — it is specific to fat handling.

42. (D) Bile acts on fat, and its action is emulsification — a physical change, with no enzyme — so the row 'Bile — fat — emulsification (physical)' is right in both columns. Answer (D). (A) is wrong: pepsin acts on protein, not starch. (B) is wrong: saliva acts on starch, not fat, and not by emulsification. (C) is wrong: bile acts on fat, not protein, and its action is physical, not chemical.

43. (D) Pancreatic juice supplies the small intestine's enzymes for carbohydrates, proteins and fats, so blocking it impairs all three together there — answer (D). (A) is wrong: salivary amylase is made in the mouth, not the pancreas. (B) is wrong: pepsin is made in the stomach, not the pancreas. (C) is wrong: water absorption is the large intestine's job and needs no pancreatic enzyme.

44. (A) Pepsin liked the stomach's strong acid, but the pancreatic (intestinal) enzymes do not; the acidic chyme must be neutralised so these enzymes can act. They do not need even stronger acid — the opposite is true. Neutralising acid is not itself the digestion of carbohydrates or fats; enzymes do that, once conditions suit them. The intestine is not kept dry; digestion needs a watery medium.

45. (C) Since no mammal's own enzymes break cellulose, a human would need the two things the cow's body supplies: cellulose-digesting microbes (as in the rumen) and the cud-re-chewing that grinds grass finer so those microbes can work — answer (C). (A) is wrong: more capacity and teeth still cannot break cellulose without microbes. (B) is wrong: saliva and acid do not chemically dissolve cellulose. (D) is wrong: a longer pipe and a gall bladder address transport and fat, not cellulose.

46. (B) The large fermentation vat where microbes break down cellulose is the rumen (Chamber 1); the chamber working like a human stomach with acid and enzymes is the abomasum (Chamber 2). Answer (B). (A) is the classic swap — exactly backwards. (C) and (D) name the reticulum and omasum, which are the other chambers and not where the human-like acid digestion mainly occurs.

47. (D) Rumination means bringing partly digested grass (the cud) back up to chew again, then re-swallowing — it is not fresh grass from outside. The re-chewed material is called cud. It is not chyme (a human stomach term). Grass cannot grow inside the stomach, so that idea is impossible. The cow is reprocessing food it already swallowed, not feeding twice.

48. (B) Losing the ability to re-chew leaves the grass in coarser pieces with less exposed surface, so the (healthy) rumen microbes can break down less of its cellulose and free fewer usable nutrients — hence less energy from the same grass. Answer (B). (A) is wrong: chewing does not heat grass enough to release energy. (C) is wrong: chewing cannot turn cellulose into protein. (D) is wrong: cellulose is not absorbed whole in the mouth.

49. (B) Cellulose breakdown keeps rising for hours after chewing stops, while the cud sits in the rumen — proof that the mouth only grinds the cud finer and the actual breakdown is done later by rumen microbes. Answer (B). (A) is wrong: if the mouth did the work, breakdown would not continue long after chewing. (C) is wrong: the food pipe only transports. (D) is wrong: the cellulose IS broken down (by microbes), not stored unchanged.

50. (A) A ruminant is defined by swallowing plant food fast and later regurgitating it as cud to chew again — that is the true reason cows, deer and buffalo are grouped together. They are plant-eaters, not meat-eaters. They do not digest food in a single cell (that describes Amoeba). They have four stomach chambers, not one, so the single-chamber claim is false.

51. (D) The bubble-like pocket is a food vacuole; when it fuses with the cell surface and dumps the undigested leftover outside, that removal is egestion. Answer (D). (A) is wrong: Amoeba has no gall bladder, and this is undigested food (egestion), not cell-made waste (excretion). (B) is wrong: a villus is a human structure, and the leftover is removed, not absorbed. (C) is wrong: a rumen is a ruminant chamber, and the leftover is discarded, not assimilated.

52. (A) Both the Amoeba's food vacuole and the human stomach and intestine are enclosed spaces where enzymes act on food OUTSIDE the body's own working cells and outside the blood — so the human match is the hollow gut cavity. Answer (A). (B) is wrong: the blood carries already-absorbed nutrients; enzymes do not digest food there. (C) is wrong: muscle cells USE nutrients (assimilation), they do not digest food. (D) is wrong: humans do not digest food on the skin.

53. (C) Forming a pseudopodium only when needed, then withdrawing it, shows that a single cell makes structures on demand for different jobs instead of maintaining fixed, specialised organs as a multi-organ human does. Answer (C). (A) is wrong: 'temporary' contradicts a permanent organ for each step. (B) is wrong: Amoeba digests inside the food vacuole, not outside its body. (D) is wrong: it certainly moves — pseudopodia serve both movement and feeding.

54. (A) The Amoeba does perform all five steps inside its single cell (using pseudopodia and a food vacuole), while the human uses many organs — so both complete the five steps, differing only in body plan. The Amoeba does digest (inside the food vacuole), so it is not mere storage. It does egest its leftovers, so it completes all five, not four. Humans certainly assimilate (they grow and repair using nutrients), so the last option is false.

55. (A) The structures the poison disables are the mitochondria, where glucose is burnt to release energy; using absorbed glucose this way is part of assimilation. Answer (A). (B) is wrong: villi are where nutrients are absorbed, not burnt, and absorption is an earlier step. (C) is wrong: food vacuoles are Amoeba's digestion pockets, and burning glucose is not digestion. (D) is wrong: salivary glands make saliva, and ingestion is taking food in, not using it.

56. (C) Protein digestion begins in the stomach (pepsin and acid), is completed in the small intestine (pancreatic and intestinal enzymes), and the resulting building blocks (amino acids) are absorbed through the villi of the small intestine. Protein digestion does not begin in the mouth (saliva acts on starch), and absorption is not in the large intestine. It does not begin in the small intestine or end in the mouth — that reverses the path. The large-intestine-then-mouth option is nonsensical for protein.

57. (A) Salivary amylase begins starch digestion in the mouth, while fat is mainly emulsified by bile and then digested by enzymes in the small intestine. They do not both begin in the stomach — starch began earlier, in the mouth, and major fat digestion waits for the small intestine. Starch does not first begin in the small intestine (the mouth starts it), and fat is not digested in the mouth. The large intestine recovers water; it is not where these foods are digested.

58. (C) The drink's dissolved sugar is a nutrient taken up mainly across the villi of the small intestine, while most remaining water is reabsorbed later in the large intestine — so the two happen in different organs. Answer (C). (A) is wrong: the stomach mainly churns and starts protein digestion, not absorption. (B) reverses the two organs. (D) is wrong: the mouth absorbs neither the sugar nor most of the water.

59. (C) Engulfing, sipping and whole-swallowing are all different ways of TAKING FOOD IN, so they differ in ingestion, not digestion. The student confuses the entry of food with its chemical breakdown. They do not differ in absorption — that internal step is not what these feeding acts show. They are not three kinds of digestion; they are one step, ingestion. Assimilation is the cellular use of nutrients, far downstream of how food is captured.

60. (A) At the instant milk enters the mouth, no step has yet been completed; the act in progress is ingestion (taking food in), the first step. Digestion has not yet been completed at that first moment, so absorption cannot be occurring. Neither ingestion nor digestion is finished at the very moment of entry, so assimilation cannot be underway. Four steps certainly have not happened — the milk has only just been taken in, with egestion (the last step) far away.